

Experimental Setup

Does the unfitted A^4 response explain the inspected Test 7 residual and predict a new locked radial waveform and local marker record?

Experiment: gross.reception-order4.v1 Model: signal-space.ss-ocf-1.flat-neutral-clock.v1
Runtime configuration: fixtures/research/gross-test-07-order4.json
Plan lock: b86e456899a69ba40670889b0702ac4f6cf47f2db9025b8d6f7610c2a8a81bf6

MODEL AND NUMERICAL SETUP

Action: Unchanged SS OCF 1 discrete radial Hamiltonian. Formal Taylor coefficients neutral odd A, A^3 and core/clock even A^0, A^2, A^4 ; exact quadrature phase series; no fitted coefficient.
Units: $c=\hbar=m=1$, r and t in m^{-1} , phase divided by 2π is cycles.
Domain: Frozen Test 6 base/fine modes at $R=80$, $h=.1, .05, .025$ (finest interpolated profile); local $r=.1$; upstream $r=14$.
Boundaries: Regular origin and zero outer radial fields. $T=65$; earliest boundary reflection cannot reach local probe.
Initial Data: Held-out center 22, width 3.25, $A=+.004, -.004, +.006$; neutral field $b=r*a$ starts exterior to surface, $b_t=b_r$; fixed clock phase 0 and amplitude .001.
Solver: RK4 $dt=.01, .01, .005$ plus independent $dt=.005$ at coarse; output .05 and decimated .1 interpolation control; optical travel-time inversion with known exterior Z , with vacuum inversion saved as control.
Fourth-order formal ODE integrated on quiet history. Prior Test 7 evidence is diagnostic only.
Tolerances: Fresh held-out record and marker criteria specified below; the prior Test 7 failures and thresholds stay fixed. First-rise/last-fall local $|a|=1e-4$, quiet-phase counterfactual.

COMPETING PREDICTIONS

controlled: Derived fourth-order sign and A^4 amplitude behavior predict the fresh held-out phase and interval within predeclared numerical and marker tolerances.
inadequate: Residual persists after fourth-order term, temporal interpolation and known exterior Z control, or markers fail; response extension remains insufficient.

CONTROLS

frozen: Exclude coefficient fits or changed action; method: Hash frozen Test 6 calibration, saved Test 7 diagnostic inputs and all predictions before new receiver evolution.
numeric: Estimate spatial, time and output-interpolation effects; method: $h=.1, .05, .025$, dt -halved coarse, output .05 decimated .1, linear versus local-derivative Hermite crossings.
reversal: Test even clock response; method: New held-out $+A$ and $-A$ plus larger amplitude .006.

PREREGISTERED DECISIONS

prediction-first (prediction-first)
Pass: Frozen calibration and inspected-history inputs verify by hash; all new surface and Taylor forecasts locked before any held-out receiver start.

Experimental Setup

Fail: Resolved finite evidence violates locked inequality.
Unresolved: Missing events or signal below five-budget resolution; no promotion.
Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

record-resolution (record-resolution)
Pass: Held-out finest $A=.004$ local interval exceeds five times $|finer-fine|+|base-time|+1e-11$ cycles, else unresolved.
Fail: Resolved finite evidence violates locked inequality.
Unresolved: Missing events or signal below five-budget resolution; no promotion.
Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

heldout-interval (heldout-interval)
Pass: Held-out $A=.004$ fourth-order interval error $\leq \max(5\% \text{ of actual interval, direct numerical budget})$; no relaxation of old Test 7 thresholds.
Fail: Resolved finite evidence violates locked inequality.
Unresolved: Missing events or signal below five-budget resolution; no promotion.
Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

heldout-waveform (heldout-waveform)
Pass: Held-out $A=.004$ fourth-order matched phase L2 error $< \min(5\%, \text{second-order L2 error})$.
Fail: Resolved finite evidence violates locked inequality.
Unresolved: Missing events or signal below five-budget resolution; no promotion.
Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

marker-timing (marker-timing)
Pass: All nonzero cases and grids have first rising and last falling $|a|=1e-4$ events; standalone forecast within 0.1 m^{-1} on each grid.
Fail: Resolved finite evidence violates locked inequality.
Unresolved: Missing events or signal below five-budget resolution; no promotion.
Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

sign-even (sign-even)
Pass: $A=+.004$ and $-.004$ give first/last marker times equal and interval differences $<1e-10$ cycles on finest grid.
Fail: Resolved finite evidence violates locked inequality.

Experimental Setup

Unresolved: Missing events or signal below five-budget resolution; no promotion.
Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

conservation (conservation)

Pass: All grid/case relative charge errors $<1e-5$; discrete total energy residual $<1\%$ of initial incident energy.

Fail: Resolved finite evidence violates locked inequality.

Unresolved: Missing events or signal below five-budget resolution; no promotion.

Uncertainty: Direct finer/fine plus base/time interval difference and $1e-11$ cycles; marker timing absolute 0.1 m^{-1} .

VISUAL QUESTIONS

prior-diagnostic [diagnostic]: Does the derived fourth-order coefficient explain the inspected residual sign and scaling?

View: PNG/SVG/PDF with exact CSV/JSON; observables: local phase, local neutral marker, numerical error

controlled: derived predicted scaling and resolved held-out agreement

inadequate: persistent residual or failed timing

Controls: frozen, numeric, reversal; uncertainty: direct grid/time differences, sample interpolation and wave inversion

Data: plot-data/summary.json; transforms: Matched quiet subtraction; exact formal A^4 quadrature expansion; no fitted coefficient.

heldout-phase [evidentiary]: Does the locked fourth-order prediction improve a new local clock history?

View: PNG/SVG/PDF with exact CSV/JSON; observables: local phase, local neutral marker, numerical error

controlled: derived predicted scaling and resolved held-out agreement

inadequate: persistent residual or failed timing

Controls: frozen, numeric, reversal; uncertainty: direct grid/time differences, sample interpolation and wave inversion

Data: plot-data/traces.csv, plot-data/summary.json; transforms: Matched quiet subtraction; exact formal A^4 quadrature expansion; no fitted coefficient.

marker-reconstruction [evidentiary]: Do derivative-informed markers and twice-frequent output explain local timing error?

View: PNG/SVG/PDF with exact CSV/JSON; observables: local phase, local neutral marker, numerical error

controlled: derived predicted scaling and resolved held-out agreement

inadequate: persistent residual or failed timing

Controls: frozen, numeric, reversal; uncertainty: direct grid/time differences, sample interpolation and wave inversion

Data: plot-data/traces.csv, plot-data/summary.json; transforms: Matched quiet subtraction; exact formal

Experimental Setup

A⁴ quadrature expansion; no fitted coefficient.

RESOURCE LIMITS

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{"max_memory_mb": 1024, "max_output_mb": 250, "max_wall_seconds": 4000}
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PROHIBITED INFERENCE

Inspected prior Test 7 data cannot support a fresh prediction claim.

Test 7 original scientific fail remains unchanged; Test 8 is blocked until separate acceptance.

No two-object recoil, invariance, gravity, angular stability or emergent spacetime inference.